SECJ3553 ARTIFICIAL INTELLIGENCE

Section 03

PROPOSAL (3%)

Theme: Agriculture

AI-Powered Predictive Analytics for Optimised Crop Yield and Risk Mitigation

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INTRODUCTION

1.0 Problem

Unpredictable weather patterns such as constant rainfall, droughts and sudden floods in the agricultural sector can greatly interfere with the planting and harvesting time. Farmers have a tendency of experiencing significant losses in their yields since they do not make predictions using data but make manual estimations or use traditional methods.

Bad weather could lead to delayed harvesting and damage crops, whereas early harvesting to avoid bad weather may lead to low yield and quality. In the absence of an intelligent decision-support system, farmers find it hard to decide the most appropriate time to plant, the most favourable harvesting period and yield with different weather conditions.

This issue is relevant to small and large-scale farmers, which results in decreased stability of income and crop management inefficiency. Therefore, an AI-based forecasting system, which can examine the environmental variables, which include temperature, humidity and rainfall to produce precise and viable agricultural data is necessary.

1.1 AI Solution

In order to solve this issue, our team wants to build a Smart Agriculture Forecasting System that is based on Artificial Intelligence to predict:

- The ideal weather and soil humidity pattern of planting.
- The most ideal time of harvest based on the predicted rainfall and environmental factors.
- The anticipated output in case it is harvested in particular periods.

The system will combine machine learning models that have been trained on previous weather and yield information. It will also have a dashboard visualisation interface which allow farmers to see predictions and recommendations clearly. Such a system will utilise AI methods such as:

Knowledge Representation	processing weather, soil and crop data into structured representations to reason.
Space Search	search through the possible planting and harvesting state to identify the best solutions.
Decision Support	applying predictive analytics and inference rules to produce decisions and recommendations.

1.2 Goal of AI Solution

The Smart Agriculture Forecasting System is primarily aimed at helping farmers make data-driven and informed decisions in agriculture.

The main goals include:

1. The time when plants should be planted and when they should be harvested in order to get maximum yield and minimise losses.
2. Offering a more interactive dashboard that displays weather, rainfall and humidity data and AI predictions.
3. Using AI reasoning to suggest the most appropriate harvesting decision, example early harvest because of a forecasted rain.
4. Promoting sustainable agriculture by using effective time and resource management.

Through these objectives, the Smart Agriculture Forecasting System will allow farmers to have a tool that will boost their productivity to grow more and removes the guesswork caused by bad weather.

PROCESS OF EMPHASIS IN DESIGN THINKING

1. WHO ARE WE EMPHATHIZING WITH

The user is an average farmer who grows crops like paddy and vegetables.

They are highly dependant on whether conditions to plant and harvest crops.

They struggle with unpredictable rainfall patterns.

2. WHAT DO THEY NEED TO DO

Farmers need to decide the optimal time to plant and harvest crops based on whether patterns.

Need to decide whether to harvest early or wait for optimal maturity of crop.

Success means reduced crop loss, higher yields and better financial stability.

3. WHAT DO THEY SEE?

The farmers see that dark clouds forming but cannot predict if it will rain heaviliy or pass.

They watch weather forecasts but find them too general and not specific enough.

4. WHAT DO THEY SAY?

The farmers say that the unpredictable weather is destroying their harvest.

They say that some technology is too expensive or complicated.

5. WHAT DO THEY DO?

Farmers check the sky multiple times daily for weather signs.

They rely on traditional indicators like wind direction and animal behavior

6. WHAT DO THEY HEAR?

Farmers constantly talk about how the weather is destroying their harvest.

Their family members express worry over financial losses from failed crops.

7. WHAT DO THEY THINK AND FEEL?

PAINS	GAINS
Losing crops due to unpredictable floods.	Reliable guidance on planting and harvesting
Financial instability to suppport family	Maximize crop yields and reduce loss
Unpredictability causes constant anxiety about when to harvest	More profitable farming to support family and invest in their farms.

PROCESS OF DEFINE IN DESIGN THINKING

USER	NEED	INSIGHT
Small to large scale farmers whose crop yield and income are heavily depended on weather conditions and accurate scheduling of planting and harvesting.	**Need 1:** To determine the optimal time to plant crops. The farmer has identified a critical pain point in starting the crop cycle and is finding a way to avoid losses from poor timing like droughts or floods right after planting.	**Insight 1:** What is the safest and best time to plant my crops? Weather and soil moisture forecast are necessary because the farmer needs to schedule planting and harvesting, also to prepare the labour and equipment.
	Need 2: To identify the best harvest time to maximise yield and quality. The farmer wanted an accurate forecast to protect the mature crops from damage like heavy rain or floods that could ruin the harvest.	**Insight 2:** Is there any bad weather like continuous rain or a flood predicted coming soon that could ruin my harvest? The farmer is worrying for its plants and needing a service that provides real-time alerts and predictive warnings to inform harvesting decisions.
	Need 3: To receive advice on managing crops. The farmer wants to minimise uncertainty and financial loss and needs a system that provides clear recommendations, not just the guessing.	**Insight 3:** What will my yield be if I harvest now versus waiting 5 more days? The farmer needs advice that offers recommendation based on data and the most profitable decision.